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Course Introduction

1) course details

Course Name: Introduction to the Computer networking
Learning platform: Simplilearn
Duration: 3hrs
Date of completion: 9th october 2026

Introduction to the course.

Computer networks is a field of computer science that deals with connecting computers and other devices to share data and resources. Network are widely used in schools, college, offices, hospitals and businesses.

This course introduces the basic concepts of networking. It covers different types of networks, Ethernet, networks topology, Ip addressing, the OSI model, the TCP/IP model, network security and network address Translation (NAT).

By completing this course, I gained a basic understanding of how networks work and how devices communicate over the internet.

Major topics covered

1) Types of networks:

I learned about PAN, LAN, MAN, and WAN
These networks differ in their coverage area and purpose.

2) Ethernet:

Ethernet is a technology used to connect devices in a wired local area network. It helps devices exchange data through network cables.

3) Network topology:

Network topology describes how devices are connected.
The main types are

1) mesh topology
2) Bus topology
3) Ring topology
4) Star topology
5) hybrid topology

4) OSI and TCP/IP models:

The OSI model has seven layers, while the TCP/IP model commonly has four layers.

These models explain how data travels from one device to another.

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OSI model.

physical -> Data link -> network -> Transport -> session -> presentation -> application.

TCP / IP model.

Physical -> Data link -> network -> Transport -> Application.

5) Ip Addressing and subnetting:

An IP addrss identifies a devices network interface interfaces a devices network interface.

IPv4 and IPv6 are the two main versions of IP. Subnetting divide a network into smaller network.

6) Network devices:
learned about routers, switches, hubs and modems. Routers connect different networks, while switches connect devices within a local network.

7) Routing and Switching:
Routing determines the path taken by data packets between networks. Switching helps transfer data between devices within a network.

8) Network protocols:
protocols are rules that allow devices to communicate Examples

include TCP, UDP, IP, HTTP, HTTPS, DNS, and DHCP.

9) Network security,
Network security protects data and devices from unauthorized access and cyberattacks. Firewalls, Encryption and authentication are important security measures.

10) Network Address Translation (NAT)
NAT translate IP address and allows multiple devices in a private network to share a public IPv4 address when accessing the internet.

Overall learning Experience:

Overall, this course was a useful learning Experience. It helped me understand the basic concepts of computer networking in a structured manner. I gained knowledge of how devices communicate, how data travels across networks and why network security is important. I would like to explore these concepts further through practical Exercises.

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CERTIFICATE OF COMPLETION

Vikas.D

has successfully completed the online course:

Introduction to the Computer Networking

This professional has demonstrated initiative and a commitment to deepening their skills and advancing their career. Well done!

9th October 2026

Certificate code : 10862435

Krishna Kumar

CEO, Simplilearn